

IPL Match Analytics Report

This report presents an end-to-end exploratory data analysis of IPL ball-by-ball match data. The project investigates how toss outcomes, innings phases, wickets, and dot balls influence match results in T20 cricket.

Project Objectives

- Analyze whether winning the toss impacts match outcomes.
- Identify which innings phase contributes most to winning.
- Compare wicket-loss patterns between winning and losing teams.
- Analyze the impact of dot balls in T20 cricket.
- Identify top-performing batters and bowlers from 2021–2025.

Data Cleaning & Preprocessing

- Removed tie and no-result matches from the analysis.
- Excluded super-over innings for consistency.
- Standardized inconsistent team names across seasons.
- Standardized season formatting into numeric values.
- Handled missing values in wicket-related columns.
- Created a match-level dataset for toss analysis.

Feature Engineering

- Created innings phase categories: Powerplay, Middle Overs, and Death Overs.
- Created toss-win indicator features.
- Created match winner indicators.
- Created dot-ball and wicket indicator features.

Key Findings

Question	Finding
Does winning the toss help?	Only slightly — toss winners won 51.5% of matches.
Most impactful phase	Middle overs (6–14) showed the biggest scoring gap.
Wicket pattern	Losing teams lost more wickets, especially in death overs.
Dot-ball impact	Losing teams played significantly more dot balls.
Top batter (2021–2025)	Shubman Gill — 2,927 runs
Top bowler (2021–2025)	Harshal Patel — 104 wickets

Most Interesting Insight

I expected the death overs to have the greatest impact on match outcomes, but the analysis revealed that the middle overs created the largest scoring gap between winning and losing teams.

Conclusion

The analysis revealed that winning the toss has only a minor impact on IPL match outcomes, whereas consistent batting performance across all innings phases — especially during the middle overs — plays a much more significant role in determining victory. Winning teams also maintained better momentum by losing fewer wickets and playing fewer dot balls throughout the innings. Additionally, players like Shubman Gill and Harshal Patel demonstrated exceptional consistency across recent IPL seasons.

Author

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